

PEREIRA, Colombia

WMO: 802100

Lat: 4.813N

Lon: 75.740W

Elev: 4416

StdP: 12.50

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	62.3	62.7	55.8	78.2	71.8	57.6	83.6	69.7	12.9	70.8	11.5	70.4	3.9	90	0.227

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.0	86.2	70.0	84.5	70.2	83.1	70.1	72.8	81.1	72.0	80.7	71.5	80.2	6.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	131.2	76.7	69.8	129.3	76.4	68.6	124.0	75.6	39.7	81.9	39.1	81.2	38.4	80.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.0	9.3	DB	59.8	89.6	1.3	2.7	58.9	91.5	58.1	93.1	57.4	94.6	56.4	96.5	(4)
			WB	56.6	74.7	1.5	1.7	55.5	75.9	54.7	77.0	53.9	77.9	52.8	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.6	73.1	73.5	72.9	72.4	72.2	72.5	72.8	73.1	73.0	71.7	71.5	72.5
	DBStd	2.20	2.31	2.22	2.45	2.25	2.03	2.02	2.03	2.13	2.13	1.95	1.83	2.04
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	8241	715	658	709	671	688	675	706	716	690	672	644	697
	CDD65	2766	250	238	244	221	223	225	241	251	240	207	194	232
	CDH74	10794	1157	1082	980	766	712	829	1015	1150	1077	644	537	845
	CDH80	1585	224	213	157	99	71	97	141	206	204	48	28	97
Wind	WSAvg	4.4	4.4	4.4	4.4	4.2	4.3	4.3	4.2	4.4	4.4	4.5	4.2	4.6
Precipitation	PrecAvg	94.4	5.5	6.5	9.0	10.6	9.6	6.3	5.1	5.0	6.4	11.7	11.3	7.5
	PrecMax	125.9	11.1	12.1	13.4	16.3	16.2	10.3	9.6	10.4	11.9	17.0	18.2	13.4
	PrecMin	65.5	2.0	2.8	4.0	5.9	5.5	1.1	1.3	1.0	3.4	6.1	7.4	1.2
	PrecStd	15.2	2.4	2.6	2.6	2.5	2.4	2.4	2.1	2.0	1.8	2.6	3.0	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.1	87.6	86.4	85.4	84.4	85.8	86.0	87.5	87.4	83.4	82.4	84.5
		MCWB	69.9	70.2	71.2	71.8	71.8	69.5	69.6	68.5	69.4	70.6	70.8	70.5
	2%	DB	84.5	84.5	83.9	82.6	82.2	82.4	83.0	84.4	84.4	81.0	80.7	82.4
		MCWB	69.8	69.5	70.7	71.2	71.4	70.6	69.9	69.6	69.7	69.7	69.9	70.3
	5%	DB	82.5	82.6	82.0	80.6	80.3	80.6	80.9	82.2	82.4	79.2	79.0	80.7
		MCWB	69.5	69.6	70.3	70.4	70.6	70.2	69.7	69.7	69.7	69.2	69.5	69.8
	10%	DB	80.6	80.7	79.3	78.6	78.5	78.7	79.1	80.3	80.3	77.6	77.2	78.9
		MCWB	68.9	69.2	69.5	69.7	69.8	69.8	69.3	69.3	69.3	68.7	68.8	69.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	72.8	73.2	73.4	73.5	73.6	72.5	72.1	72.2	72.2	72.3	72.6
		MCDB	82.8	83.0	81.9	81.2	81.3	80.2	81.2	82.0	82.1	80.4	79.4	80.5
	2%	WB	70.9	71.6	71.8	72.1	72.1	71.9	71.3	71.0	71.0	70.8	70.9	71.3
		MCDB	81.3	81.7	80.7	80.0	79.6	79.6	80.0	80.9	81.2	79.0	78.4	79.6
	5%	WB	70.1	70.6	70.9	71.1	71.0	70.9	70.2	70.2	70.2	69.9	69.9	70.5
		MCDB	80.0	79.9	79.4	78.7	78.2	78.5	78.8	79.8	80.0	77.6	77.4	78.5
	10%	WB	69.3	69.7	70.0	70.0	70.0	70.0	69.6	69.4	69.4	69.1	69.1	69.6
		MCDB	78.5	78.3	78.0	77.1	76.9	77.2	78.0	78.4	78.6	76.2	76.0	77.4
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.0	15.1	14.3	13.7	14.4	15.5	16.1	16.4	14.5	14.1	14.8
		MCDBR	19.0	18.3	17.8	16.9	16.0	16.5	17.5	18.8	19.3	16.2	15.6	16.7
		MCWBR	8.3	7.7	7.6	8.1	7.9	7.9	8.0	8.2	8.6	7.7	7.4	7.8
	5% WB	MCDBR	16.8	16.4	16.1	15.5	14.8	15.4	16.4	17.2	17.5	15.5	15.0	15.6
	MCWBR	7.8	7.3	7.6	7.9	7.8	7.8	8.0	7.9	8.2	8.2	7.7	7.4	7.7
Clear-Sky Solar Irradiance	taub	0.425	0.458	0.485	0.449	0.404	0.381	0.384	0.412	0.424	0.436	0.417	0.415	
	taud	2.317	2.216	2.140	2.263	2.406	2.475	2.461	2.358	2.316	2.299	2.364	2.365	
	Ebn at Noon	284	277	269	273	280	283	283	280	281	279	284	285	
	Edh at Noon	42	47	51	44	37	34	35	40	43	43	40	39	
All-Sky Solar Radiation	RadAvg	1380	1377	1326	1331	1374	1411	1473	1482	1481	1372	1307	1313	
	RadStd	107	122	87	68	59	54	68	74	80	56	57	81	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)		+0.61	+0.58	N/A	N/A	N/A	N/A	N/A	-333	+331	N/A

Nomenclature: See separate page